Magnetic Force on Moving Charges (completed)

Goal.

- See why a charge moving across a uniform $\vec{B}$ field goes in a circle, and where the cyclotron radius $r = mv/|q|B$ comes from.
- Work a full numerical example end to end: accelerate a charge through a potential difference, then find the field strength from how fast it circles (Example 29.11).
- Split $\vec{v}$ into the piece along $\vec{B}$ (untouched) and the piece across it (circles), to see why a charge entering at an angle traces a *helix*, not a circle.
- Find both the radius of that helix and its *pitch* — the distance the charge drifts along $\vec{B}$ during one full loop.

1. Why a uniform field makes a charge go in a circle

A charge q moves with speed v perpendicular to a uniform field $\vec{B}$. The magnetic force $\vec{F} = q\vec{v} \times \vec{B}$ is always perpendicular to $\vec{v}$ — it only ever turns the velocity, never speeds it up or slows it down. A force that's always perpendicular to the motion, with constant magnitude, is exactly what produces uniform circular motion.

Set the magnetic force equal to the centripetal force it's providing:

$$|q|\, v\, B \;=\; \frac{mv^2}{r}$$

Solve for the radius of the circle:

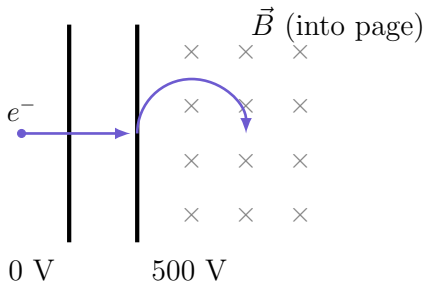
$r_{\text{cyc}} = \dfrac{mv}{|q|B}$

Check. Does a *faster* charge circle in a bigger or smaller radius? Does a *stronger* field give a bigger or smaller radius? Is either answer the opposite of what you expected?

Faster ⇒ bigger radius ($r \propto v$). Stronger field ⇒ smaller radius ($r \propto 1/B$) — a stronger field turns the charge *more sharply*, not less. That second one trips people up if they picture a stronger field as a "bigger push" that would fling the charge outward; instead it's a tighter leash.

2. Worked example: the radius of cyclotron motion

Example 29.11. An electron is accelerated from rest through a potential difference of 500 *V, then injected into a uniform magnetic field. Once in the field, it completes half a revolution in* 2.0 *ns. What is the radius of its orbit?*



Label the 0 V and 500 V plates, mark $\vec{B}$'s direction, and sketch the electron's path: straight across the gap, then curving once it enters the field.

Step 1 — speed entering the field. Energy conservation, $K_f + qV_f = K_i + qV_i$, with $K_i = 0$:

$$\tfrac{1}{2}mv_f^2 = e\,V_f \qquad \Longrightarrow \qquad v_f = \sqrt{\frac{2eV_f}{m}}$$

$v_f = 1.33 \times 10^7$ m/s

Step 2 — turn the half-revolution time into a period and frequency. Half a revolution takes 2.0 ns, so a full revolution (the period T) takes 4.0 ns. The cyclotron frequency is $f = 1/T = 2.5 \times 10^8$ Hz.

Step 3 — use the cyclotron frequency to get B. The cyclotron frequency doesn't depend on v or r at all:

$$f = \frac{|q|B}{2\pi m} \qquad \Longrightarrow \qquad B = \frac{2\pi m f}{e}$$

$B = 8.94 \times 10^{-3}$ T

Step 4 — now find the radius, using v_f from Step 1 and B from Step 3:

$r_{\mathrm{cyc}} = \dfrac{mv_f}{eB} = 8.5 \times 10^{-3}\ \mathbf{m} = 8.5\ \mathbf{mm}$

Check. The review in the book compares this radius to a photo of an electron beam curving in a magnetic field. Does 8.5 mm sound like a reasonable size for an electron's orbit in a lab-scale field, or suspiciously large/small?

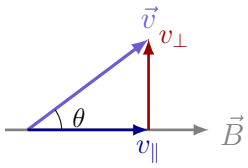
Reasonable — a 17-mm-*diameter* orbit is the kind of thing you can actually see curve in a tabletop electron-beam tube. If this had come out at, say, 8.5 *nm* or 8.5 km, that would be the signal to go back and check a power of ten.

3. When $\vec{v}$ isn't perpendicular to $\vec{B}$: helical motion

Every example so far has had $\vec{v} \perp \vec{B}$ exactly. What if the charge enters at some other angle θ to the field?

Split $\vec{v}$ into two pieces, one along $\vec{B}$ and one across it:

$$\vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp}\,, \qquad v_{\parallel} = v\cos\theta\,, \qquad v_{\perp} = v\sin\theta$$



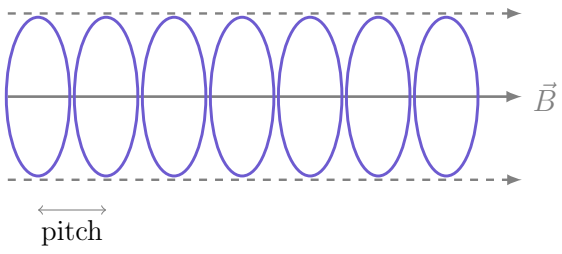
- The parallel piece, $\vec{v}_{\parallel}$, is parallel (or anti-parallel) to $\vec{B}$ — so $\vec{v}_{\parallel} \times \vec{B} = 0$: zero force at all. Left alone, it just drifts at constant velocity along $\vec{B}$, forever.
- The perpendicular piece, $v_{\perp}$, is exactly §1's situation all over again — it circles with radius $r = \dfrac{mv_{\perp}}{|q|B}$ (the §1 formula, with $v_{\perp}$ in place of v).

Put the two motions together — a uniform drift along $\vec{B}$, plus a circle in the plane across $\vec{B}$ — and the path traced out is a helix: a corkscrew / spring-shaped path that advances steadily along $\vec{B}$ while circling around it.

Check. At $\theta = 90°$ you're back to §1's pure circle. What does the path look like in the other extreme, $\theta = 0°$?

At $\theta = 0°$, $v_{\perp} = 0$ — there's no circling at all, and the charge just travels in a straight line along $\vec{B}$ at constant speed v. A straight line is the $\theta \to 0$ limit of a helix with the pitch stretched out to infinity (zero radius).

New problem. A proton ($m = 1.67 \times 10^{-27}$ kg, $q = +e = 1.60 \times 10^{-19}$ C) enters a region of uniform magnetic field $B = 0.50$ T. Its velocity has magnitude $v = 4.0 \times 10^{5}$ m/s and points 30° away from $\vec{B}$. Find (a) the radius of the helix it traces, and (b) the pitch *— the distance it drifts along $\vec{B}$ during one full revolution.*



Mark one full turn of the helix and label the pitch you're solving for.

Step 1 — components of v relative to $\vec{B}$:

$v_\parallel = v\cos 30° = 3.46 \times 10^5$ m/s $\qquad$ $v_\perp = v\sin 30° = 2.0 \times 10^5$ m/s

Step 2 — radius of the helix (same formula as §1, using $v_\perp$):

$r_{\text{helix}} = \frac{mv_\perp}{qB} = 4.2 \times 10^{-3}$ **m** $= 4.2$ **mm**

Step 3 — period of one full revolution (this doesn't depend on $v_\perp$ at all — why not? see §1):

$$T = \frac{2\pi m}{qB} = 1.31 \times 10^{-7} \text{ s} = 131 \text{ ns}$$

$T = 1.31 \times 10^{-7}$ s

Step 4 — the pitch. During one period T, the parallel motion just moves at constant velocity $v_\parallel$:

pitch $= v_\parallel T = 4.5 \times 10^{-2}$ **m** $= 4.5$ **cm**

Check. If the proton had instead entered at a *smaller* angle θ (closer to parallel with $\vec{B}$), would the helix get fatter or skinnier? Would it get more tightly or more loosely wound (bigger or smaller pitch)?

Smaller θ means less of v is perpendicular ($v_\perp = v\sin\theta$ shrinks), so the helix gets *skinnier* (smaller radius). At the same time more of v is parallel ($v_\parallel = v\cos\theta$ grows), so the pitch grows too — the helix gets *more loosely wound.* (The period T itself doesn't change at all — it never depended on θ.)